PHAN VAN BANG

AI Engineer — Multi-Agent Systems, RAG, Voice AI · Da Nang, Vietnam

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SUMMARY

AI Engineer with 3+ years building production LLM systems at MekongAI: multi-agent orchestration, RAG/GraphRAG, voice AI and IoT, delivered end to end (FastAPI, React, Docker, AWS) across 8+ business domains. Author of the open-source Nexus multi-agent platform (190+ MCP tools, 218 test files).

TECHNICAL SKILLS

AI / LLM	Multi-agent orchestration, RAG & GraphRAG (vector + full-text + graph, RRF), MCP (client & server, OAuth 2.1 PKCE), LangGraph, LangChain, LlamaIndex, CrewAI, LoRA/QLoRA fine-tuning (Unsloth), vLLM
Voice & Vision	Whisper STT, TTS (OpenAI, Fish Audio, ElevenLabs), VAD, PaddleOCR, SAM3, ComfyUI / Stable Diffusion, spaCy (vi/en)
Backend	Python, TypeScript, FastAPI, SQLAlchemy + Alembic, Celery, WebSocket, SSE, MQTT, REST
Data	PostgreSQL + pgvector, Qdrant, MongoDB, Redis, Neo4j
Frontend	React, Next.js, Zustand, Tailwind CSS
DevOps	Docker Compose, AWS (EC2, S3), GitHub Actions, Nginx, RunPod (GPU serverless), Pytest
Also	C#, Java, Go, Playwright, n8n

WORK EXPERIENCE

AI Engineer · MekongAI — 06/2023 – Present

- Architected **MekongAI Nexus**, a multi-agent orchestration platform (FastAPI, React 19, PostgreSQL + pgvector, Redis) with 190+ MCP tools and structured agent debate (challenge, rebuttal, synthesis); serves 500+ conversations/day at 85%+ autonomous resolution.
- Built a hybrid RAG pipeline (vector + full-text + graph with RRF fusion) with GraphRAG memory: 92% answer accuracy over 10,000+ documents and +35% retrieval accuracy versus a vector-only baseline.
- Designed layered agent memory (L0–L3) and an XSkill continual-learning loop (skills, experiences, trajectory compression into SFT data), plus a background memory-consolidation job.
- Hardened the agent runtime for production: circuit breaker and provider fallback across 8+ LLM backends, task checkpoint and auto-resume, command / SSRF / exfiltration guards, secret redaction, and 218 automated test files.
- Shipped **Knowlympic**: PaddleOCR pipeline plus LLM question generation that cut content creation time by 80% for 3 English language centers.
- Built **MetaHuman** voice platform (full-duplex Whisper STT, streaming TTS, VAD) at <500 ms response latency; connected ESP32 voice bots to the same backend via MCP and MQTT.
- Delivered computer-vision and GenAI services: 20+ ComfyUI workflows (5,000+ images/month), a face-similarity API, and SAM3 / PaddleOCR-VL on RunPod GPU serverless.
- Fine-tuned Llama 3.1 (8B/70B) with Unsloth (LoRA/QLoRA) on trajectory-compressed SFT data; delivered 56+ AI projects across Legal, Education, Recruitment, POS, Healthcare, Crypto, Social Media and Content Generation.

SELECTED PROJECTS

MekongAI Nexus – open-source multi-agent platform — github.com/pvbang/nexus

- 25 REST endpoint modules, 68 MCP tool modules, OpenAI-compatible API, SSE streaming, 9 chat channels (Telegram, Discord, Slack, WhatsApp, Email, ...).
- Docker Compose deployment, 19 Alembic migrations, React 19 UI with live process cards for agent reasoning, debate and tool calls.

Python 3.12, FastAPI, React 19, TypeScript, PostgreSQL + pgvector, Redis, LiteLLM, Docker

AI Vehicle-Repair Assistant (client project)

- LangGraph agent with a 5-layer pipeline (reasoning, search, evaluation, answer, verification) answering multi-DTC diagnostic queries with cited references, streamed over SSE.
- One-command bootstrap for the full stack; MongoDB backup to S3 with SHA-256 verification and staging-first restore.

LangGraph, FastAPI, MongoDB, Qdrant, Redis, Streamlit, AWS S3, Docker

AI Recruitment Platform – Vietnamese NLP matching

- CV parsing with OCR and a dual spaCy pipeline (vi + en); job-candidate matching by vector similarity.
  FastAPI, Qdrant, spaCy, OCR, AWS S3

M-Auto – AI YouTube video studio

- Idea-to-upload pipeline: LLM scripts, image and TTS generation, FFmpeg rendering (Ken Burns, xfade, karaoke captions) with an AI quality critic that re-runs weak stages.
  Python, Streamlit, FFmpeg, Edge-TTS, faster-whisper

B. Nexus Quantum – MetaTrader 5 trading EA (commercial) b-nexus-quantum.vercel.app

- 11-module MQL5 Expert Advisor: 21 entry modes, 18-layer weighted signal scoring, multi-layer risk controls, licensing; sold with a public landing page.

Open-source developer tools github.com/pvbang

- mcp-prune: strips cosmetic fields from MCP tool schemas, cutting tokens by 73.8% on the bundled fixture while validating the same inputs.
- sentinel-scan: zero-dependency static scanner for JS/TS (hardcoded secrets, eval/exec sinks, weak crypto).

EDUCATION & LANGUAGES

B.Eng. in Information Technology · Vietnam-Korea University of Information and Communication Technology (VKU), Da Nang 2020 – 2025

- Languages: Vietnamese (native), English (professional working proficiency: technical reading and writing).